

Quick Activity 2: Canteen Group Order

Problem Analysis

Inputs	Process	Outputs
Number of rice meals	Calculate meal subtotal	Meal subtotal
Number of drinks	Calculate drink subtotal	Drink subtotal
Amount paid	Calculate 10% discount on meals	Discount
	Calculate packaging fee	Packaging fee
	Calculate final bill	Final bill
	Calculate change	Change

Pseudo Code

START

INPUT riceMeals

INPUT drinks

INPUT amountPaid

$\text{mealSubtotal} = \text{riceMeals} * 85$

$\text{drinkSubtotal} = \text{drinks} * 25$

$\text{discount} = \text{mealSubtotal} * 0.10$

$\text{packagingFee} = \text{riceMeals} * 5$

$\text{finalBill} = \text{mealSubtotal} - \text{discount} + \text{drinkSubtotal} + \text{packagingFee}$

$\text{change} = \text{amountPaid} - \text{finalBill}$

OUTPUT mealSubtotal

OUTPUT drinkSubtotal

OUTPUT discount
OUTPUT packagingFee
OUTPUT finalBill
OUTPUT change

END

Dry-run Calculations TEST CASE 1: 3 Rice Meals, 4 Drinks, ₱500.00 Payment

Step	Variable	Calculation	Result
1.	amountPaid	500.00	₱500.00
2.	mealSubtotal	3×85.00	₱255.00
3.	drinkSubtotal	4×25.00	₱100.00
4.	discount	$\text{mealSubtotal} \times 0.10$	₱25.50
5.	packagingFee	3×5.00	₱15.00
6.	finalBill	$\text{mealSubtotal} - \text{discount} + \text{drinkSubtotal} + \text{packagingFee}$	₱344.50
7.	change	$\text{amountPaid} - \text{finalBill}$	₱155.50

Dry-run Calculations TEST CASE 2: 5 Rice Meals, 3 Drinks, ₱600.00 Payment

Step	Variable	Calculation	Result
1.	amountPaid	600.00	₱600.00
2.	mealSubtotal	5×85.00	₱425.00
3.	drinkSubtotal	3×25.00	₱75.00
4.	discount	$\text{mealSubtotal} \times 0.10$	₱42.50
5.	packagingFee	5×5.00	₱25.00
6.	finalBill	$\text{mealSubtotal} - \text{discount} + \text{drinkSubtotal} + \text{packagingFee}$	₱482.50
7.	change	$\text{amountPaid} - \text{finalBill}$	₱117.50